What our laboratory affords

capability profiles, design requirements and their matching

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Approach

A laboratory has devices; a device bears *functions*, dispositions realized only in processes of one kind. A laboratory's capability classes are defined by those functions, by the environments it provides and by its validated readout bindings, so HermiT classifies laboratories from their devices. A design's requirement classes are defined by its content (administrations, readouts, environment factors), so HermiT classifies designs by their needs. Matching a design's requirements, reagents, readout bindings and timing against the laboratories of a site is a closed-world question over the individuals. Cell 01's profile is read from its ten component records, the device registry and the twin's durations; the core plate-reader facility is ILLUSTRATIVE.

Axioms

Laboratories, devices and capabilities

Laboratory ⊑ MaterialEntity ⊓ ∃*hasDevice*.Device

hasDevice ⊑ *hasPart*

Device ⊑ MaterialEntity

Disjoint(Laboratory, Device, OrganoidCulture, ChemicalAgent, Plate, Well)

DeviceFunction ⊑ Disposition ⊓ ∃*inheresIn*.Device

DosingFunction ⊑ DeviceFunction ⊓ ∀*realizedIn*.Administration

IncubationFunction ⊑ DeviceFunction ⊓ ∀*realizedIn*.Incubation

BrightfieldImagingFunction ⊑ DeviceFunction ⊓ ∀*realizedIn*.ImagingAssay

LuminescenceReadingFunction ⊑ DeviceFunction ⊓ ∀*realizedIn*.LuminescenceAssay

AbsorbanceReadingFunction ⊑ DeviceFunction ⊓ ∀*realizedIn*.AbsorbanceAssay

CalibrationFunction ⊑ DeviceFunction

QuarantineFunction ⊑ DeviceFunction

TransportFunction ⊑ DeviceFunction

PlateHandlingFunction ⊑ DeviceFunction

StorageFunction ⊑ DeviceFunction

EndpointAssay ⊑ Assay

LuminescenceAssay ⊑ EndpointAssay

AbsorbanceAssay ⊑ EndpointAssay

ImagingAssay ⊓ EndpointAssay ⊑ ⊥

MetabolicActivity ⊑ Quality ⊓ ∃*inheresIn*.OrganoidCulture

EnvironmentProfile ⊑ PlanSpecification

ReadoutBinding ⊑ InformationContentEntity ⊓ ∃*isAbout*.Quality

ValidatedReadoutBinding ⊑ ReadoutBinding

DosingLab ≡ Laboratory ⊓ ∃*hasDevice*.∃*bearerOf* .DosingFunction

CultivationLab ≡ Laboratory ⊓ ∃*hasDevice*.∃*bearerOf* .IncubationFunction

BrightfieldImagingLab ≡ Laboratory ⊓ ∃*hasDevice*.∃*bearerOf* .BrightfieldImagingFunction

LuminescenceReadoutLab ≡ Laboratory ⊓ ∃*hasDevice*.∃*bearerOf* .LuminescenceReadingFunction

AbsorbanceReadoutLab ≡ Laboratory ⊓ ∃*hasDevice*.∃*bearerOf* .AbsorbanceReadingFunction

QualityControlledLab ≡ Laboratory ⊓ ∃*hasDevice*.∃*bearerOf* .CalibrationFunction

ContainmentCapableLab ≡ Laboratory ⊓ ∃*hasDevice*.∃*bearerOf* .QuarantineFunction

ConnectedLab ≡ Laboratory ⊓ ∃*hasDevice*.∃*bearerOf* .TransportFunction

MultiEnvironmentLab ≡ Laboratory ⊓ (≥ 2*providesEnvironment*.EnvironmentProfile)

ValidatedSizeReadoutLab ≡ Laboratory ⊓ ∃*hasReadoutBinding*.(ValidatedReadoutBinding ⊓ ∃*isAbout*.OrganoidSize)

ValidatedMetabolicReadoutLab ≡ Laboratory ⊓ ∃*hasReadoutBinding*.(ValidatedReadoutBinding ⊓ ∃*isAbout*.MetabolicActivity)

TreatmentAndImagingModule ≡ DosingLab ⊓ CultivationLab ⊓ BrightfieldImagingLab ⊓ ValidatedSizeReadoutLab

Design requirements

RequiresDosing ≡ StudyDesign ⊓ ∃*hasArm*.∃*hasAdministrationSpec*.AdministrationSpecification

RequiresCultivation ≡ StudyDesign ⊓ ∃*hasArm*.ArmSpecification

ImagingReadout ⊑ ReadoutSpecification

EndpointReadout ⊑ ReadoutSpecification

LuminescenceReadout ⊑ EndpointReadout

AbsorbanceReadout ⊑ EndpointReadout

ImagingReadout ⊓ EndpointReadout ⊑ ⊥

RequiresImaging ≡ StudyDesign ⊓ ∃*hasReadout*.ImagingReadout

RequiresLuminescence ≡ StudyDesign ⊓ ∃*hasReadout*.LuminescenceReadout

RequiresAbsorbance ≡ StudyDesign ⊓ ∃*hasReadout*.AbsorbanceReadout

RequiresEndpointHandling ≡ StudyDesign ⊓ ∃*hasReadout*.EndpointReadout

EnvironmentFactor ⊑ Factor

RequiresMultipleEnvironments ≡ StudyDesign ⊓ ∃*hasFactor*.EnvironmentFactor

Laboratories

Cell 01 (automated treatment-and-imaging module)

device	capability verbs (component record)	functions
H4 Quarantine Station	isolate_sample, retain_invalid_run, prevent_reentry, release_by_authorization	QuarantineFunction
L09 Automated Plate Hotel	buffer_plate, store_plate, retrieve_plate, report_inventory, reserve_slot, verify_identity	StorageFunction
L10 Automated CO2 Incubator	incubate_plate, maintain_environment, store_plate, release_plate, report_environment, raise_excursion_alarm	IncubationFunction, StorageFunction
L12 Automated Liquid-Handling Station	pick_tip, detect_liquid_surface, aspirate, dispense, mix, dose_well, eject_tip	DosingFunction
L17 Bright-Field Imaging Station	load_plate, autofocus, image_well, image_plate_region, compute_focus_quality, export_raw_image	BrightfieldImagingFunction
L21 Quality-Control and Calibration Station	run_reference_measurement, validate_imager, validate_barcode, validate_tool_frame	CalibrationFunction
R01 Fixed Precision Plate-Handling Arm	pick_sbs_plate, place_sbs_plate, handover_plate, reseat_plate, inspect_grasp, move_plate_between_local_stations	PlateHandlingFunction
R02 Rail-Mounted Flexible Manipulation System	translate_workspace, pick_plate, pick_tube, handle_rack, press_button, turn_selector, open_access_door, change_tool, inspect_station	—
R03 Clean Sample AMR with H2 Deterministic Dock	navigate, transport_clean_sample, dock_precisely, handover_carrier, opportunity_charge	TransportFunction
R02 Automatic Tool Changer and Tool Magazine	dock_tool, lock_tool, verify_tool, unlock_tool, store_tool, select_tool_for_task	—

Environments: standard 'temperature_C': 37.0, 'CO2_percent': 5.0, 'humidity_percent': 90.0. Reagent stocks: challenge, rescue. Validated readouts: organoid_radius_um. Dosing within the 600 s excursion window: at most 11 wells per session. **Capability classes (HermiT):** BrightfieldImagingLab, ConnectedLab, ContainmentCapableLab, CultivationLab, DosingLab, QualityControlledLab, TreatmentAndImagingModule, ValidatedSizeReadoutLab.

Core facility: multimode plate reader (ILLUSTRATIVE)

device	capability verbs (component record)	functions
Multimode plate reader	read_luminescence, read_absorbance, pick_sbs_plate	AbsorbanceReadingFunction, LuminescenceReadingFunction, PlateHandlingFunction
AMR reception dock	receive_carrier	TransportFunction

Environments: none. Reagent stocks: atp_assay_reagent. Validated readouts: atp_luminescence. **Capability classes (HermiT):** AbsorbanceReadoutLab, ConnectedLab, LuminescenceReadoutLab, ValidatedMetabolicReadoutLab.

Designs

The current challenge-rescue design

Design classes: FactorialDesign, HypothesisDiscriminatingDesign, PreSpecifiedDesign, SufficientlyInformativeDesign, ValidatedInjuryRescueDesign, WellContrastedDesign. Requirements (HermiT): RequiresCultivation, RequiresDosing, RequiresImaging. **Verdict: executable in cell01**.

requirement	needs	offered by	allocated
RequiresCultivation	CultivationLab	cell01	cell01
RequiresDosing	DosingLab	cell01	cell01
RequiresImaging	BrightfieldImagingLab	cell01	cell01
reagents stocked	challenge, rescue	cell01	cell01
validated binding: imaging readout	OrganoidSize	cell01	cell01
units within the plate's interior wells	3 of 60	cell01	cell01

Notes:

- doses per time point 0.0: 2, 48.0: 1; at most 11 wells per session (600 s window, 205 s transfer overhead, 35.6 s per well); one session per time point

The improved design (vehicle, rescue-only, early rescue, 3 replicates)

Design classes: AdequatelyControlledInjuryRescueDesign, DeviationAwareDesign, FactorialDesign, HypothesisDiscriminatingDesign, InteractionAwareDesign, SufficientlyInformativeDesign, TherapeuticWindowDesign. Requirements (HermiT): RequiresCultivation, RequiresDosing, RequiresImaging. **Verdict: not executable as modelled**.

requirement	needs	offered by	allocated
RequiresCultivation	CultivationLab	cell01	cell01
RequiresDosing	DosingLab	cell01	cell01
RequiresImaging	BrightfieldImagingLab	cell01	cell01
reagents stocked	challenge, rescue, vehicle	cell01	unmet
validated binding: imaging readout	OrganoidSize	cell01	cell01
units within the plate's interior wells	15 of 60	cell01	cell01

Missing:

- reagent not stocked in cell01: vehicle

With vehicle stocked in Cell 01 the verdict would be: **executable in cell01**. Notes:

- doses per time point 0: 15, 6: 15, 48: 15; at most 11 wells per session (600 s window, 205 s transfer overhead, 35.6 s per well); split into several sessions at 0 h (2), 6 h (2), 48 h (2); the last session of a time point starts about 10 min after the first

The improved design with an ATP luminescence endpoint at 72 h

Design classes: AdequatelyControlledInjuryRescueDesign, DeviationAwareDesign, FactorialDesign, HypothesisDiscriminatingDesign, InteractionAwareDesign, SufficientlyInformativeDesign, TherapeuticWindowDesign. Requirements (HermiT): RequiresCultivation, RequiresDosing, RequiresEndpointHandling, RequiresImaging, RequiresLuminescence. **Verdict: not executable as modelled**.

requirement	needs	offered by	allocated
RequiresCultivation	CultivationLab	cell01	cell01
RequiresDosing	DosingLab	cell01	cell01
RequiresImaging	BrightfieldImagingLab	cell01	cell01
RequiresLuminescence	LuminescenceReadoutLab	core_reader	core_reader
reagents stocked	challenge, rescue, vehicle	cell01	unmet
validated binding: imaging readout	OrganoidSize	cell01	cell01
validated binding: luminescence readout	MetabolicActivity	core_reader	core_reader
units within the plate's interior wells	15 of 60	cell01	cell01

Missing:

- reagent not stocked in cell01: vehicle

With vehicle stocked in Cell 01 the verdict would be: **executable on site with transfers**. Notes:

- doses per time point 0: 15, 6: 15, 48: 15; at most 11 wells per session (600 s window, 205 s transfer overhead, 35.6 s per well); split into several sessions at 0 h (2), 6 h (2), 48 h (2); the last session of a time point starts about 10 min after the first
- transfer cell01 → core_reader: possible (clean-sample transport at both ends); the plate leaves the controlled environment for an unmodelled time

A challenge x oxygen design (environment factor)

Design classes: ControlledDesign, DeviationAwareDesign, FactorialDesign, HypothesisDiscriminatingDesign, InteractionAwareDesign, ReplicatedDesign, WellContrastedDesign. Requirements (HermiT): RequiresCultivation, RequiresDosing, RequiresImaging, RequiresMultipleEnvironments. **Verdict: not executable as modelled**.

requirement	needs	offered by	allocated
RequiresCultivation	CultivationLab	cell01	cell01
RequiresDosing	DosingLab	cell01	cell01
RequiresImaging	BrightfieldImagingLab	cell01	cell01
RequiresMultipleEnvironments	MultiEnvironmentLab	—	unmet
reagents stocked	challenge	cell01	cell01
validated binding: imaging readout	OrganoidSize	cell01	cell01
units within the plate's interior wells	12 of 60	cell01	cell01

Missing:

- RequiresMultipleEnvironments: no laboratory is a MultiEnvironmentLab

Notes:

- environment factor 'oxygen' needs 2 controlled environments; provided: cell01 1, core_reader 0
- doses per time point 0: 6; at most 11 wells per session (600 s window, 205 s transfer overhead, 35.6 s per well); one session per time point